PRE-ALGEBRA (7TH) · UNIT 7: SOLVING EQUATIONS AND INEQUALITIES

Summer Job Savings

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students choose a summer job to buy a laptop. They solve two-step and multi-step equations for earnings, compare jobs with variables on both sides, use two-step inequalities for goals, and compare simple and compound interest.

Rounding decisions ("at least" vs. "at most") come directly from the context.

Standards Addressed

- 7.EE.B.3 Solve multi-step problems with rational numbers
- 7.EE.B.4a Solve $px + q = r$ and $p(x + q) = r$
- 7.EE.B.4b Solve and interpret $px + q > r$
- 8.EE.C.7 Variables on both sides; simple and compound interest

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch: "What would you buy with a summer job?" Part 1 earnings equations.
Day 2	Part 2 savings inequalities.
Day 3	Part 3 interest and the summer plan.

Materials

- Student handout and Job Offer Cards
- Calculators
- Optional: bank interest rate ads

How the Three Versions Differ

Level	What changes
CP2	Two-step equations and inequalities with starters, a number-line graph, and one simple-interest calculation.
CP1	Variables on both sides, distributive equations, weekly constraints, transforming I = Prt, and bank comparison.

Level	What changes
Honors	Common-unit job models, break-even points, multiple constraints, mixed plans, and long-term interest tables.

Watch For (Common Misconceptions)

- Dividing before subtracting in two-step equations.
- Rounding down when the goal requires "at least".
- Using 3 instead of 0.03 for 3%.

Extension Ideas

- Research real summer job pay in your town.
- Graph earnings for two jobs on the same axes.
- Add sales tax to the laptop price.

Answer Key – CP2

Part 1: Earn the Money

#	Task	Answer
1	Mowing: write an expression for the money earned from m lawns after paying for the $40 tune-up.	15m − 40
2	How many lawns must you mow to earn $200 after the tune-up? Solve 15m − 40 = 200.	15m = 240 → m = 16 lawns
3	Babysitting: solve 12h − 20 = 172 to find the hours needed.	12h = 192 → h = 16 hours
4	Check your mowing answer by substituting.	15(16) − 40 = 240 − 40 = 200 ✓

Part 2: Reach the Goal

#	Task	Answer
5	Write and solve an inequality for the number of dog walks w you need: 85 + 8w ≥ 650.	8w ≥ 565 → w ≥ 70.6 → 71 walks
6	Snack stand: 85 + 25 + 10.50h ≥ 650. Solve for h.	10.50h ≥ 540 → h ≥ 51.4 → 52 hours
7	If you work 12 hours a week at the snack stand, how many weeks is that?	52 ÷ 12 ≈ 4.3 → 5 weeks
8	Graph the solution of w ≥ 71 on a number line.	Closed circle at 71, shaded right.

Part 3: Save with Interest

#	Task	Answer
9	You put $300 in Bank A at 3% for 1 year. How much interest do you earn?	I = 300 × 0.03 × 1 = $9
10	How much is in the account after 2 years with simple interest?	300 + 300(0.03)(2) = $318
11	Choose a job and write your full summer plan with an equation or inequality.	Answers vary.

Reflection

#	Prompt	Look for
R1	Why do you undo addition or subtraction first when solving a two-step equation?	It reverses the order of operations so the variable term is isolated before dividing.
R2	Why do you round 70.6 walks up to 71?	You need at least $650; 70 walks would not be enough.

Answer Key – CP1

Part 1: Compare Jobs

#	Task	Answer
1	After how many hours do babysitting and the snack stand pay the same total? Solve $12h - 20 = 10.50h + 25$.	$1.5h = 45 \rightarrow h = 30$ hours
2	Which job pays more for 20 hours? For 40 hours?	20 h: babysitting \$220, snack \$235 → snack. 40 h: babysitting \$460, snack \$445 → babysitting.
3	A mowing weekend: you mow l lawns on Saturday and 3 more on Sunday, earning $15(l + l + 3) - 40 = 365$. Solve for l.	$15(2l + 3) - 40 = 365 \rightarrow 30l + 45 = 405 \rightarrow l = 12$
4	Solve $\frac{2}{3}(h - 6) = 18$ and explain a babysitting situation it could model.	$h - 6 = 27 \rightarrow h = 33$; e.g., two-thirds of the hours after the first 6.

Part 2: Savings Goal

#	Task	Answer
5	Write and solve an inequality for the number of babysitting hours needed.	$85 + 12h - 20 \geq 650 \rightarrow 12h \geq 585 \rightarrow h \geq 48.75 \rightarrow$ 49 hours
6	You can work at most 10 hours a week at the snack stand. Write an inequality for the number of weeks to reach \$650 and solve it.	$85 + 25 + 105w \geq 650 \rightarrow 105w \geq 540 \rightarrow w \geq 5.14 \rightarrow$ 6 weeks
7	You will spend \$15 per week on snacks while working. Rewrite and solve the inequality from the last question.	$110 + 90w \geq 650 \rightarrow w \geq 6 \rightarrow$ 6 weeks
8	Explain how you knew whether to round up or down in each answer.	You need at least the goal, so round the number of hours or weeks up.

Part 3: Interest

#	Task	Answer
9	Solve $I = Prt$ for r. What rate gives \$24 interest on \$400 in 2 years?	$r = \frac{I}{Pt}$; $r = \frac{24}{800} = 0.03 = 3\%$
10	Compare \$650 for 3 years in Bank A (3% simple) and Bank B (2.8% compounded yearly).	A: $650 + 650(0.03)(3) = \$708.50$. B: $650(1.028)^3 \approx \$706.14$. Bank A earns more.
11	Write your summer plan: job(s), hours or lawns, weeks, and where to save the extra money. Support each choice with an equation or inequality.	Answers vary.

Reflection

#	Prompt	Look for
R1	What does the solution h = 30 mean when comparing two jobs?	At 30 hours both jobs pay the same; before that one is better and after it the other is.
R2	Why might compound interest beat simple interest over many years even with a lower rate?	Compound interest earns interest on interest, so it grows faster over time.

Answer Key – Honors

Part 1: Job Modeling

#	Task	Answer
1	Write total-earnings expressions for all four jobs after start-up costs, with h hours (snack, babysitting), l lawns, or w walks. Assume one lawn takes 1.25 h and one walk takes 0.5 h. Rewrite each in terms of hours.	Mow: $15(\frac{h}{1.25}) - 40 = 12h - 40$. Babysit: $12h - 20$. Snack: $10.5h + 25$. Dog: $8(\frac{h}{0.5}) = 16h$.
2	Rank the jobs by total earnings for 40 hours.	Dog 640, Babysit 460, Snack 445, Mow 440
3	When does mowing earn the same as the snack stand? Solve and interpret.	$12h - 40 = 10.5h + 25 \rightarrow 1.5h = 65 \rightarrow h \approx 43.3$ hours
4	Dog walking has the best rate, but only 20 walks per week are available. Write the most you can earn from dogs in 6 weeks and the hours used.	120 walks × 8 = $960 using 60 hours

Part 2: Constraints

#	Task	Answer
5	If you only walk dogs (max 20 walks/week), write and solve an inequality for the weeks needed.	$85 + 160w \geq 650 \rightarrow w \geq 3.53 \rightarrow 4$ weeks
6	You want to save $650 AND keep at least $150 spending money. Write a two-step inequality for babysitting hours and solve.	$85 + 12h - 20 \geq 800 \rightarrow 12h \geq 735 \rightarrow h \geq 61.25 \rightarrow 62$ hours
7	Is 62 hours possible within the 15-hour weekly limit over 8 weeks? How much of the limit is used?	$62 \leq 120$, yes; $\frac{62}{120} \approx 52\%$
8	Design a mixed plan (two jobs) that reaches $800 within 8 weeks and 15 h/week. Show the equation for your total.	Answers vary; e.g., 10 dog walks/week (5 h) + 10 h babysitting/week: $8(80 + 120) - 20 = 1{,}580 \geq 800$.

Part 3: Growing the Money

#	Task	Answer
9	Solve $A = P(1 + rt)$ for t. How long until $650 grows to $750 at 3% simple interest?	$t = \frac{A - P}{Pr}$; $\frac{100}{19.5} \approx 5.1$ years
10	Make a table for $650 in Bank A and Bank B for t = 1, 5, 10, 20 years. When does Bank B pull ahead?	A: $669.50, $747.50, $845.00, $1,040.00. B: $668.20, $746.24, $856.73, $1,129.21. Bank B pulls ahead between year 5 and year 10.
11	Write a recommendation for a younger sibling about choosing a bank for long-term savings.	Answers vary; look for the table and the compound vs. simple reasoning.

Reflection

#	Prompt	Look for
R1	Why did rewriting every job in terms of hours make comparing them easier?	It puts all plans on the same scale, so slopes (pay rates) can be compared directly.
R2	Explain why the simple and compound accounts switch which is better.	Simple interest grows by the same amount each year; compound grows by a percent of a growing balance, so it eventually wins.